

HAND GESTURE CONTROLLER

Project Setup & User Guide

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A Python-based computer vision project that uses a webcam to detect hand gestures and perform computer actions such as mouse movement, clicking, video control, volume control, scrolling, and screenshots.

Feature	Description
Real-time Hand Tracking	Detects hand landmarks through a webcam using MediaPipe.
Computer Control	Uses PyAutoGUI to perform mouse, click, volume, scrolling and media actions.
Computer Vision	Uses OpenCV for webcam capture, processing and display.
Screenshot Control	Three-finger gesture held continuously for 5 seconds captures a screenshot.

1. System Requirements

Hardware

- Laptop/Desktop
- Webcam
- Minimum 4 GB RAM
- Dual-core processor or better
- Keyboard and mouse

Software

- Windows 10/11
- Python 3.9–3.11 (Python 3.11 recommended)
- Visual Studio Code
- Webcam drivers

2. Tools & Technologies

Technology	Purpose
Python	Main programming language
OpenCV	Webcam capture and image processing
MediaPipe	Hand tracking and landmark detection
PyAutoGUI	Mouse, click, volume, scrolling and computer control
VS Code	Development environment

3. Installation

Check Python

```
python --version
```

Check pip

```
python -m pip --version
```

Install OpenCV

```
python -m pip install opencv-python
```

Install MediaPipe

```
python -m pip install mediapipe==0.10.21
```

Install PyAutoGUI

```
python -m pip install pyautogui
```

Optional virtual environment

```
python -m venv venv  
venv\Scripts\activate
```

4. Run the Project

Save the main program as **main.py** in the project folder and run it from the activated virtual environment:

```
python main.py
```

5. Gesture Controls

Gesture	Action
Open Palm	Play / Toggle Video
Fist	Pause / Toggle Video
Thumb Up	Increase Volume
Thumb Down	Decrease Volume
Index Finger	Move Mouse Cursor
Index + Thumb Pinch	Left Click
Two Fingers	Right Click
Pinky Finger UP	Scroll Up
Pinky Finger DOWN	Scroll Down
Three Fingers — hold 5 seconds	Take Screenshot
Q	Exit Program

Important: Pinky scrolling uses only the pinky finger. The three-finger gesture is reserved only for the 5-second screenshot action.

6. Screenshot Function

Show three fingers and keep the gesture stable for 5 continuous seconds. The screenshot is saved automatically inside the **screenshots** folder with a timestamped filename.

7. Scrolling

For vertical scrolling, point the pinky upward for Scroll Up and downward for Scroll Down. Keep the mouse cursor over the webpage, PDF, document, or other scrollable area because the computer-control library performs the scroll at the current cursor position.

8. Project Structure

A simple GitHub repository can use the following structure:

```
AI_Hand_Gesture_Controller/  
■■■ main.py  
■■■ test.py  
■■■ screenshots/  
■■■ README.md  
■■■ Hand_Gesture_Controller_Guide.pdf
```

9. Troubleshooting

Webcam does not open

Check that the webcam is connected and not being exclusively used by another application. If necessary, verify the camera index in the OpenCV VideoCapture call.

ModuleNotFoundError: No module named 'cv2'

Make sure the correct virtual environment is active, then run:

```
python -m pip install opencv-python
```

ModuleNotFoundError: No module named 'mediapipe'

```
python -m pip install mediapipe==0.10.21
```

Scrolling does not happen

Keep the pinky clearly extended and point it vertically upward or downward. Also place the mouse cursor over the area that should scroll.

10. GitHub Upload Checklist

Before publishing the repository, include the source code, this guide, a clear README, the requirements/install commands, and a short demonstration section. Avoid uploading personal files, virtual-environment folders, or unnecessary generated screenshots.

11. Author

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Hand Gesture Controller — Python Computer Vision Project